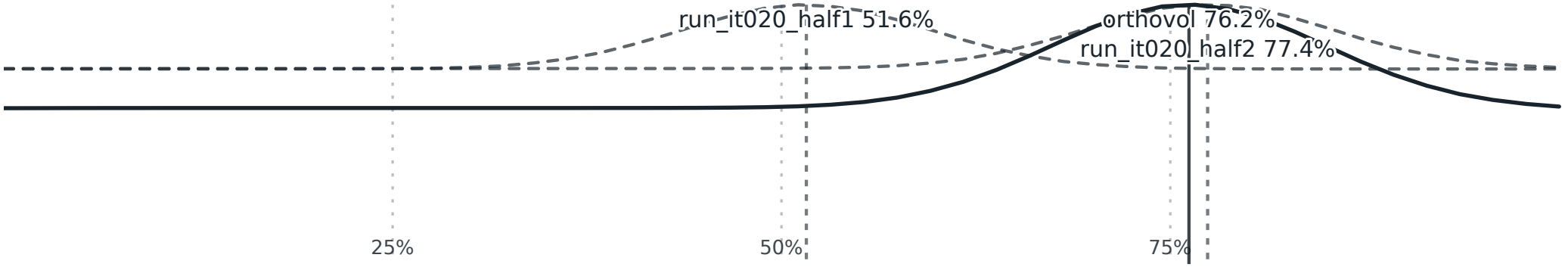
CryoFlow session QC report

Project **β-Galactosidase Tutorial (demo)** · 21 jobs — 16 succeeded · 1 running · 0 failed · 4 waiting.

Pipeline at a glance

- **16** succeeded jobs feed this report's QC sections;
- **1** running · **0** failed — failures stay counted here, never dropped;
- **4** waiting (idle or submitted) — no verdict exists for work that has not run.

Map QC



Map QC summary — orthovol

Job `cmu4tv3ji0014tjyvtp8ucgnw` · mean-density landscape along **Z** (64 bins).

- **Peak** mean ρ at plane 48 (**76.2%** of depth) — where the specimen's mass concentrates on this axis
- **Trough** at plane 0 (0.0%)
- **Span** across planes: 0.0278 (max 0.0695, min 0.0417)

Comparison maps (2)

Map	Bins	Peak at	Agreement r	Verdict	Shape
run_it020_half1	32	51.6%	-0.25	diverges	
run_it020_half2	32	77.4%	0.99	agrees	

Where a comparison line follows the main landscape, the density is consistent between maps; where it parts ways lives noise or masking. Agreement r is the Pearson correlation on the shared 0–100% fraction scale — each terrain self-scaled to its own map's stats (the shape is the signal, not absolute ρ).

Local agreement

Map	Q1 (0–25%)	Q2 (25–50%)	Q3 (50–75%)	Q4 (75–100%)	Weakest	Depth (fraction)	Bands
run_it020_half1	0.32	0.82	-0.94	0.68	Q3 (50–75%) (-0.94)	0.50–0.75	
run_it020_half2	0.26	0.99	0.99	1.00	Q1 (0–25%) (0.26)	0.00–0.25	

A global r can hide a localized betrayal — a map can agree over most of the depth and part ways in a single band (a mask edge, a noise shelf). Each quarter of the shared fraction scale is correlated independently, and the weakest quarter is the address to inspect; the Depth column quotes that address as fractions (the same numbers the viewer's bracket door jumps to). Where the half-maps disagree with each other in one band only, suspect that band, not the reconstruction.

Pairwise agreement

Map A	Map B	Agreement r	Verdict	Depth (fraction)
run_it020_half1	run_it020_half2	-0.28	diverges	0.50–0.75

Two half-maps come from disjoint halves of the data — where they agree with EACH OTHER, the density is real (this is the question FSC asks). Maps that follow the main landscape but not each other deserve a second look. The Depth column quotes where each pair's corroboration is thinnest — the same fractions the viewer's bracket doors jump to, measured between the pair itself.

Reading the depth addresses

Every depth in this report is a fraction of the map's thickness along the axis (0 = the front face, 1 = the back face), never an Å position — the landscape is mean density per plane, contour-independent, describing the whole map. The quarter bands are cut by an equal-count knife: each quarter holds the same number of measured planes, so a weak quarter names a real place, not a sparse one. The tables divide the labour — Comparison summarizes each map with one global r; Local addresses (the weakest quarter, quoted as depth); Pairwise corroborates (each pair measured as its own two maps). The Comparison table deliberately carries no Depth column: its address is already quoted in the Local agreement section below — a depth printed twice has two fathers and drifts apart. In the viewer, every one of these addresses is a door: press the chip or the bracket and the plane lands on the band's centre.

Exported from CryoFlow's slice instrument — the mean-density landscape is contour-independent and describes the whole map, not the current isosurface.

Session map inventory

Every job in this session that owns a true 3D volume, newest walk first. The deep profiles above ride the newest owner; this inventory keeps every other owner visible — no map hides below the fold (the t211 lesson: a walk that stops at the first winner leaves the rest of the world unseen). Each row is a door — press it and the page hands you to that job's results (the t210 lesson carried onto the report: a name on a roster should never be a dead end). The Peak column quotes each owner's main map exactly the way the deep report quotes the winner — where the mass concentrates along the shared axis, as a fraction of depth; — means still

measuring, never a guess. The Δ winner column reads every peak against the winner's own — the reference row speaks +0.0, its dash obeys the same law, and the lens never guesses; on the page the row farthest from the winner wears the amber edge (a lens, not a verdict — the exported bytes keep the numbers and let the reader judge). The Agreement r column gives every owner the winner's own comparison arithmetic — the same Pearson correlation on the shared 0–100% fraction scale, each owner's main landscape read against the winner's on the finer of the two grids; the reference row lands at 1.00 through the same path, and — means still measuring or a flat field, never a guess. The Weakest column answers the r column's next question — WHERE the shape does not follow: the same quarter-band cut the deep report's Local agreement table speaks, thinnest band first, Q1–Q4 with the band's own r; the reference row's thinnest quarter is perfect and the cell says so.

Job	Main map	Volumes	Peak	Δ winner	Agreement r	Weakest	Shape
QA Refine3D	orthovol	4	76.2%	+0.0	1.00	Q1 (1.00)	
QA Class2D Source	orthovol	1	25.4%	-50.8	-0.43	Q2 (-0.59)	

Scheduling sweep

No sweep raced this session — open the HPC queue panel and run *Compare profiles*; the winner's verdict will be bound here verbatim.

Bound from this session's live state — the pipeline glance, the map landscape and the sweep verdict each keep their own provenance. The filename carries the export stamp.